Photo: Movie Poster

HW 5 due this Friday, Mar 13.
HW 6 coming soon.

Quiz 7 due next Monday

Class 15: Turing Machines

University of Virginia
CS3120: DMT2
https://weikailin.github.io/cs3120-toc
Wei-Kai Lin

Plan

Midterm 1

Turing Machines

Story of Turing

Variants

Textbook [TCS] Section 7

https://introtcs.org/public/lec_06_loops.html
Loops and infinity

Midterm 1: DFA and Regular Expr.

Compared to Midterm 0: higher mean, lower stdev.

Regrade requests: open after this class.

True/False: most are good, Problem 1(e) is lowest.

(e) There exist languages $L_1, L_2 \subseteq \{0,1\}^*$ such that L_1 is regular, but L_2 is not regular, yet $L = L_1 \cup L_2$ is regular.

$\boxed{\text{T}}$ $\boxed{\text{F}}$ Reason:

$\{0,1\}^*$? $\{0,1\}^*$

Is Tamagotchi a DFA? Why?

Digital "pet"
3 buttons
320-byte RAM
Microcontroller (CPU)

Alphabet: 3+1 buttons
States: $2^{320 \times 8}$
Transitions: [C] × [4]
Accepting:

https://tamagotchi-official.com/us/history/

Recap: Turing Machine Model

A model for writing algorithms / programs

A *Turing Machine,* is defined by (Σ, C, δ):

Σ : alphabet $-$ finite set of symbols

$$\Sigma \supseteq \{0, 1, \triangleright, \emptyset\}$$

1. $C \in \mathbb{N}$: a finite number of states
2. $\boldsymbol{\delta}$: transition function

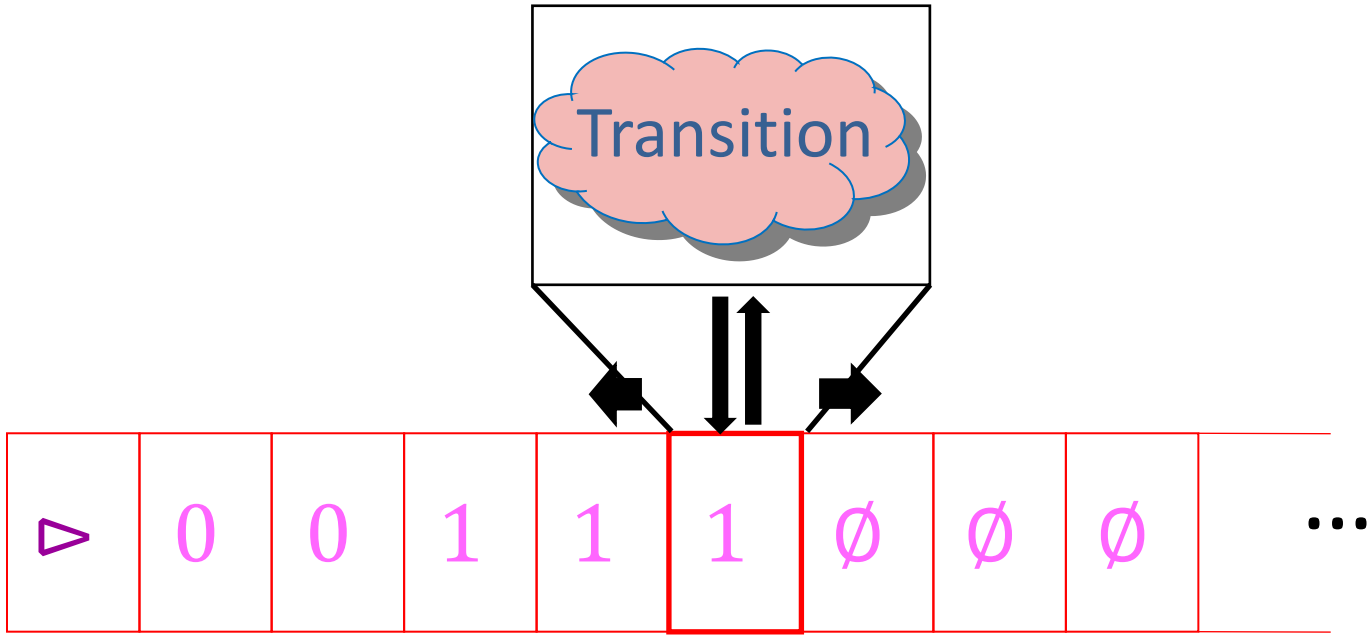
$$\boldsymbol{\delta}: [C] \times \Sigma \to [C] \times \Sigma \times \{\mathbf{L}, \mathbf{R}, \mathbf{S}, \mathbf{H}\}$$

Left, **R**ight, **S**tay, **H**alt

A **(deterministic)** ***finite automaton*** over alphabet {0,1} is a tuple (C, T, S) where:

1. C --- the number of *states*
2. $T: [C] \times \{0,1\} \to [C]$ --- a transition function
3. $S \subseteq [C]$ --- the set of accept states

Turing Machine



Tape initially contains the input followed by blanks

If transition "halts", → tape contains the output

Semi-infinite tape (memory)

Does infinite memory model any real-world computer?

Yes real comp is finite

No can model "future" machines

Registration Survey

What made you passionate about Discrete Mathematics?

Many reasons …
Simple blocks build big objects

Turing Machines

Executing Turing Machines

A *Turing Machine,* is defined by (Σ, C, δ):

$C \in \mathbb{N}$: a finite number of states

Σ : alphabet – finite set of symbols

$$\Sigma \supseteq \{0, 1, \triangleright, \emptyset\}$$

$\boldsymbol{\delta}$: transition function

$$\boldsymbol{\delta}: [C] \times \Sigma \to [C] \times \Sigma \times \{\mathbf{L}, \mathbf{R}, \mathbf{S}, \mathbf{H}\}$$

Execution of a TM

Definition. The execution of a TM, $M = (\Sigma, C, \delta)$ on input $x \in \{0,1\}^*$ is this process:

1. Initialize T as $\triangleright, x_0, x_1, \ldots, x_{n-1}, \emptyset, \emptyset, \emptyset, \ldots$ where $n = |x|$.

   That is, $T[0] = \triangleright$, $T[i+1] = x_i$ for $i \in [n]$, and $T[i] = \emptyset$ for $i > n$.
2. Initialize two natural number variables, $i = 0, s = 0$.

   initial state
3. **repeat**
   1. $(s', \sigma', D) = \boldsymbol{\delta}(s, T[i])$
   2. $s := s', T[i] := \sigma'$
   3. if $D = \mathbf{R}$: $i := i + 1$

      if $D = \mathbf{L}$: $i := \max\{i - 1, 0\}$

      if $D = \mathbf{H}$: **break**

      Halt

Output of executing TM

Definition. The **output** of the execution of a TM, $M = (\Sigma, C, \delta)$ on input $x \in \{0,1\}^*$ is the result of this process:

input

1. Initialize T as $\boxed{\triangleright, x_0, x_1, \dots, x_{n-1}, \emptyset, \emptyset, \emptyset,}$ … where $n = |x|$.

   That is, $T[0] = \triangleright$, $T[i+1] = x_i$ for $i \in [n]$, and $T[i] = \emptyset$ for $i > n$.
2. Initialize two natural number variables, $i = 0, s = 0$.
3. **repeat**
   1. $(s', \sigma', D) = \boldsymbol{\delta}(s, T[i])$
   2. $s := s', T[i] := \sigma'$
   3. if D = **R**: $i := i + 1$

      if D = **L**: $i := \max\{i - 1, 0\}$

      if D = **H**: **break**

Is the output well-defined if $M(x) \neq \perp$?

output

4. If the process finishes (the repeat breaks), the result of this process is:

   $M(x) = \boxed{T[1], \dots, T[m]}$ where $m > 0$ is the smallest integer, $T[m+1] \notin \{0,1\}$.

   Otherwise (i.e., the machine does NOT halt), then define $M(x) = \perp$. \bot

   [warning: $\perp$ is not an actual output]

Special Simpler Case: No Input Execution

Definition. The **output** of the execution of a TM, $M = (\Sigma, C, \delta)$ is the result of this process:

input x = " "

1. Initialize T as $\triangleright$, $\emptyset$, $\emptyset$, $\emptyset$, ...
2. Initialize two natural number variables, $i = 0, s = 0$.
3. **repeat**
    1. $(s', \sigma', D) = \boldsymbol{\delta}(s, T[i])$
    2. $s := s', T[i] := \sigma'$
    3. if D = $\mathbf{R}$: $i := i + 1$
       if D = $\mathbf{L}$: $i := \max\{i - 1, 0\}$
       if D = $\mathbf{H}$: **break**
4. If the process finishes (the repeat breaks), the result of this process is:
   $M(x) = T[1], \ldots, T[m]$ where $m > 0$ is the smallest integer, $T[m+1] \notin \{0,1\}$.
   Otherwise (i.e., the machine does NOT halt), then define $M(x) = \perp$.
   [warning: $\perp$ is not an actual output]

Who needs input?

Thm / Lemma

Given a TM, $M = (\Sigma, C, \delta)$ and input x, produce a TM, $M' = (\Sigma, C', \delta')$ such that the output $M'(x)$ in the No-Input model, is the same as the output $M(x)$ in the normal model.

Why always possible?

M' : TM W_x : write output x ; move tape head to $\triangleright$ location

run M

Simulating Turing Machines

http://morphett.info/turing/turing.html

```
0 0 0 r 0
0 1 1 r b
0 _ _ * halt-accept
b 0 0 r b
b 1 1 r b
b _ _ * halt-reject
```

Choosing other TM definitions

- Output before halt
- Two-way infinite
- More symbols
- No starting ⊳ symbol

- More? Stay tuned…

Story of Turing

Was this really Turing's model?

A *Turing Machine,* is defined by (Σ, C, δ):
$C \in \mathbb{N}$: a finite number of states
Σ : alphabet $-$ finite set of symbols

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$$\boldsymbol{\delta}: [C] \times \Sigma \to [C] \times \Sigma \times \{\mathbf{L}, \mathbf{R}, \mathbf{S}, \mathbf{H}\}$$

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1. Initialize T as $\triangleright, x_0, x_1, \dots, x_{n-1}, \emptyset, \emptyset, \emptyset, \dots$ where $n = |x|$.
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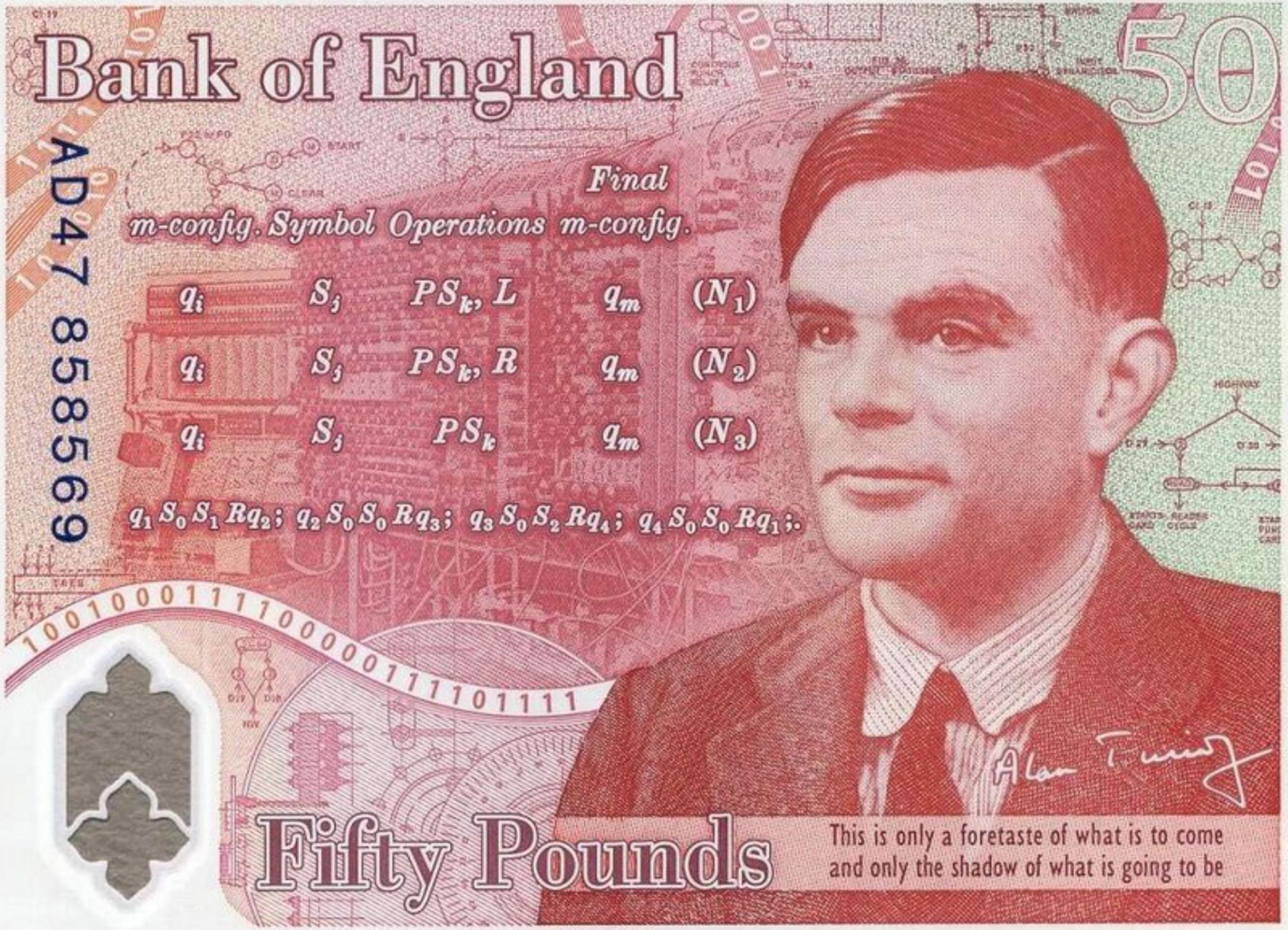
   $M(x) = T[1], \dots, T[m]$ where $m > 0$ is the smallest integer, $T[m + 1] \notin \{0,1\}$.

Otherwise (i.e., the machine does NOT halt), $M(x) = \perp$.



https://www.bankofengland.co.uk/banknotes/polymer-50-pound-note

https://blog.cloudflare.com/details-of-the-cloudflare-outage-on-july-2-2019/

From Class 16

CLOUDFLARE The Cloudflare Blog

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Email Address Subscribe

Product News | Speed & Reliability | Security | Serverless | Zero Trust | Developers | Deep Dive | Life @Cloudflare

Details of the Cloudflare outage on July 2, 2019

07/12/2019

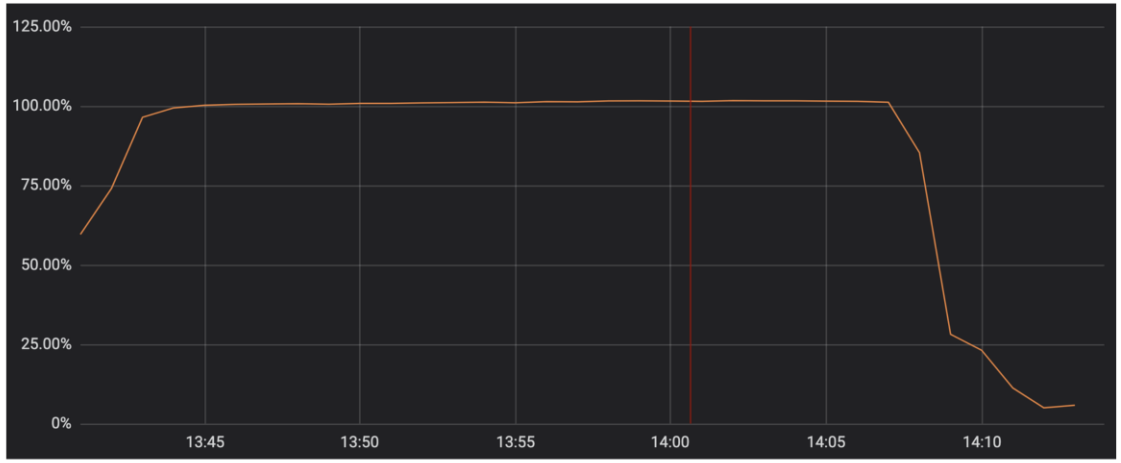
John Graham-Cumming

CTO of Cloudflare

1 alert for Global Traffic Drop

02/07/2019, 13:48 — 1 alert

Service: Escalate Edge SRE — #53967



CPU utilization in one of our PoPs during the incident

This resulted in our customers (and their customers) seeing a 502 error page when visiting any Cloudflare domain. The 502 errors were generated by the front line Cloudflare web servers that still had CPU cores available but were unable to reach the processes that serve HTTP/HTTPS traffic.

502 Bad Gateway

cloudflare

80% of Cloudflare's customers traffic dropped due to excessive cost of matching a regular expression!

John Graham-Cumming's blog

https://blog.jgc.org/2009/06/alan-turing-deserves-apology-from.html

2009-06-23

Alan Turing deserves an apology from the British Government

When I started writing The Geek Atlas there was one name that was getting in the book no matter what: Alan Turing.

Alan Turing matters on many levels because he was, in the words of the memorial in Manchester:

> Father of computer science, mathematician, logician, wartime codebreaker, victim of prejudice

Turing's work has affected us all. He's best know for his involvement in Second World War code breaking (especially for helping to break Engima) and if all he had done was that we would be grateful.

But Turing was also a critical pioneer of computer science. He defined a theoretical model of computers (at a time when 'computer' meant a person, often a woman, who computed numbers) that holds true today. He suggested how we might determine whether a computer was sentient (with the Turing Test).

Turing's death should remind us how prejudice ruins and degrades.

Alan Turing was gay. And he was prosecuted for 'indecent acts' and eventually took his own life aged 41. This man, younger than me, killed himself because at the time homosexuality was illegal and having been prosecuted he was chemically castrated in an attempt to 'cure' him. He had been stripped of his security clearance.

For years, his legacy was largely ignored outside the computer community. To quote Wikipedia:

> In 1994 a stretch of the A6010 road (the Manchester city intermediate ring road) was named Alan Turing Way. A bridge carrying this road was widened, and carries the name 'Alan Turing Bridge'.

A frikkin' Ring Road!

It wasn't until 2001 that a statue was erected.

John Graham-Cumming's blog

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A frikkin' Ring Road!

It wasn't until 2001 that a statue was erected.

Today is Alan Turing's 97th birthday. Or at least it could have been if it were not for his prosecution and untimely death.

Isn't it time the British Government apologized for the way he was treated? We shouldn't let this anniversary of his death go by without recognizing the great works this man did and the ignominious way in which he was treated.

at June 23, 2009

Labels: alan turing, rants and raves

https://web.archive.org/web/20100301233733/http://petitions.number10.gov.uk/turing/

Number10.gov.uk BETA

The official site of the Prime Minister's Office

10

We the undersigned petition the Prime Minister to apologize for the prosecution of Alan Turing that led to his untimely death. More details

Submitted by John Graham-Cumming – **Deadline to sign up by:** 20 January 2010 – **Signatures:** 32,280

More details from petition creator

Alan Turing was the greatest computer scientist ever born in Britain. He laid the foundations of computing, helped break the Nazi Enigma code and told us how to tell whether a machine could think.

He was also gay. He was prosecuted for being gay, chemically castrated as a 'cure', and took his own life, aged 41.

The British Government should apologize to Alan Turing for his treatment and recognize that his work created much of the world we live in and saved us from Nazi Germany. And an apology would recognize the tragic consequences of prejudice that ended this man's life and career.

https://web.archive.org/web/20100301233733/http://petitions.number10.gov.uk/turing/

Number10.gov.uk BETA

The official site of the Prime Minister's Office

10

Petition update, 11 September 2009, while petition was still open

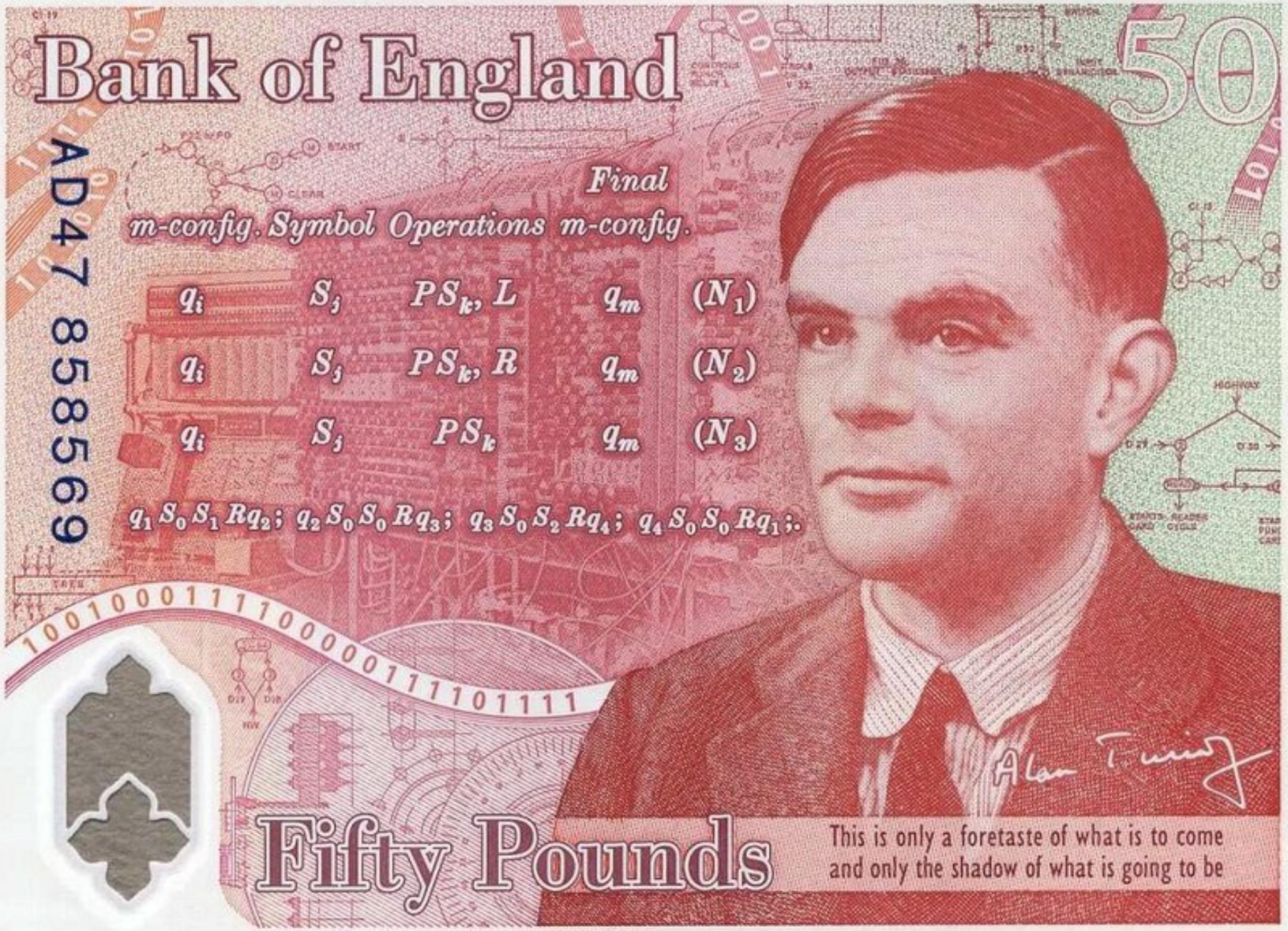
Thank you for signing this petition. The Prime Minister has written a response. Please read below.

Prime Minister: 2009 has been a year of deep reflection – a chance for Britain, as a nation, to commemorate the profound debts we owe to those who came before. A unique combination of anniversaries and events have stirred in us that sense of pride and gratitude which characterise the British experience. Earlier this year I stood with Presidents Sarkozy and Obama to honour the service and the sacrifice of the heroes who stormed the beaches of Normandy 65 years ago. And just last week, we marked the 70 years which have passed since the British government declared its willingness to take up arms against Fascism and declared the outbreak of World War Two. So I am both pleased and proud that, thanks to a coalition of computer scientists, historians and LGBT activists, we have this year a chance to mark and celebrate another contribution to Britain's fight against the darkness of dictatorship; that of code-breaker Alan Turing.

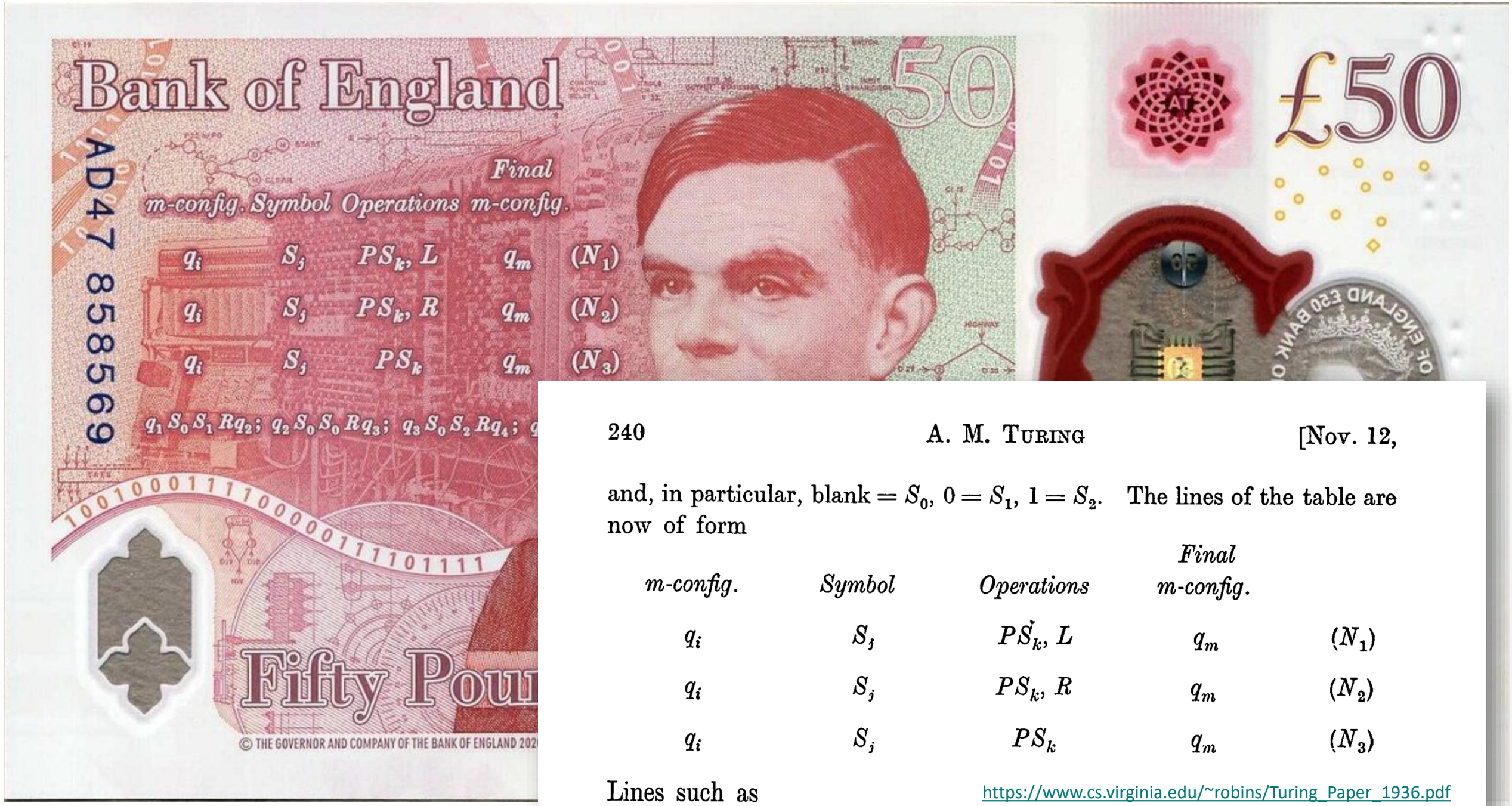
But even more than that, Alan deserves recognition for his contribution to humankind. For those of us born after 1945, into a Europe which is united, democratic and at peace, it is hard to imagine that our continent was once the theatre of mankind's darkest hour. It is difficult to believe that in living memory, people could become so consumed by hate – by anti-Semitism, by homophobia, by xenophobia and other murderous prejudices – that the gas chambers and crematoria became a piece of the European landscape as surely as the galleries and universities and concert halls which had marked out the European civilisation for hundreds of years. It is thanks to men and women who were totally committed to fighting fascism, people like Alan Turing, that the horrors of the Holocaust and of total war are part of Europe's history and not Europe's present.

So on behalf of the British government, and all those who live freely thanks to Alan's work I am very proud to say: we're sorry, you deserved so much better.

Gordon Brown


Bank of England
AD47 858569
Final
m-config. Symbol Operations m-config.
q_i S_j PS_k, L q_m (N_1)
q_i S_j PS_k, R q_m (N_2)
q_i S_j PS_k q_m (N_3)
$q_1 S_0 S_1 R q_2;\ q_2 S_0 S_0 R q_3;\ q_3 S_0 S_2 R q_4;\ q_4 S_0 S_0 R q_1;.$
1001000111100001111101111
Fifty Pounds
This is only a foretaste of what is to come
and only the shadow of what is going to be
Alan Turing 1912-1954
© THE GOVERNOR AND COMPANY OF THE BANK OF ENGLAND 2020

£50
BANK OF ENGLAND £50
AD47 858569



and, in particular, blank $= S_0$, $0 = S_1$, $1 = S_2$. The lines of the table are now of form

m-config.	*Symbol*	*Operations*	*Final m-config.*	
q_i	S_j	PS_k, L	q_m	(N_1)
q_i	S_j	PS_k, R	q_m	(N_2)
q_i	S_j	PS_k	q_m	(N_3)

Lines such as

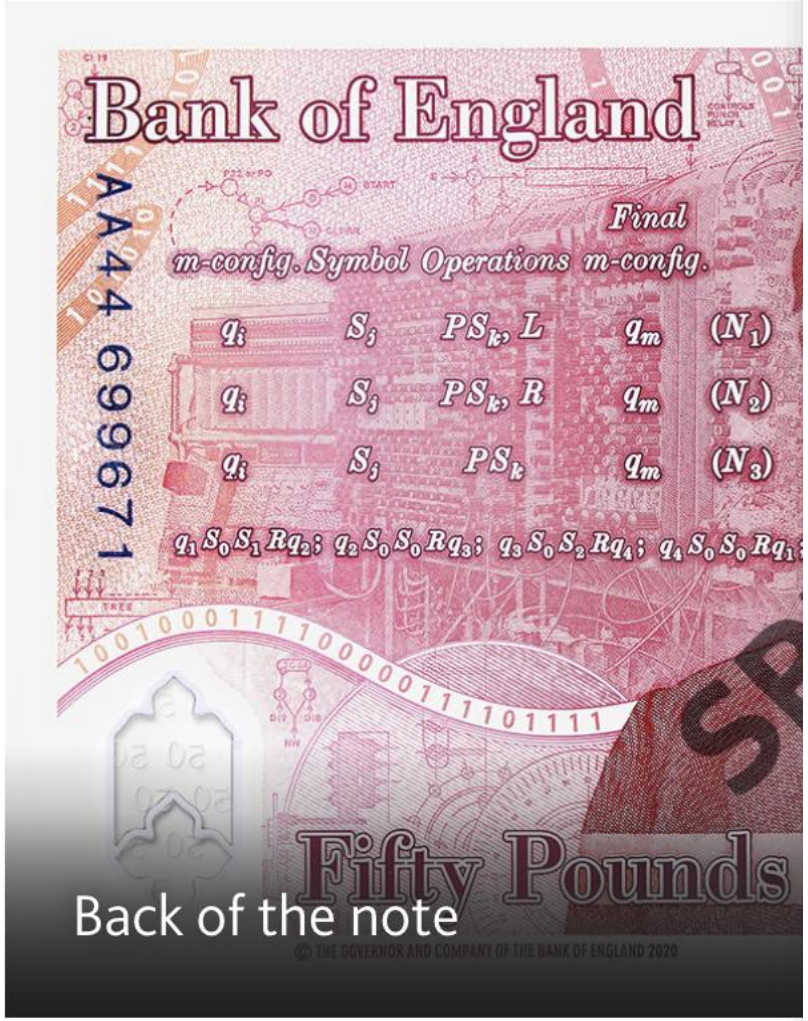
https://www.cs.virginia.edu/~robins/Turing_Paper_1936.pdf

ON COMPUTABLE NUMBERS, WITH AN APPLICATION TO THE ENTSCHEIDUNGSPROBLEM

By A. M. TURING.

[Received 28 May, 1936.—Read 12 November, 1936.]

The "computable" numbers may be described briefly as the real numbers whose expressions as a decimal are calculable by finite means. Although the subject of this paper is ostensibly the computable *numbers*, it is almost equally easy to define and investigate computable functions of an integral variable or a real or computable variable, computable



Let us find a description number for the machine I of § 3. When we rename the m-configurations its table becomes:

q_1	S_0	PS_1, R	q_2
q_2	S_0	PS_0, R	q_3
q_3	S_0	PS_2, R	q_4
q_4	S_0	PS_0, R	q_1

Other tables could be obtained by adding irrelevant lines such as

q_1	S_1	PS_1, R	q_2

Our first standard form would be

$$q_1 S_0 S_1 R q_2;\ q_2 S_0 S_0 R q_3;\ q_3 S_0 S_2 R q_4;\ q_4 S_0 S_0 R q_1;.$$

The standard description is

$$DADDCRDAA;DAADDRDAAA;$$

$$DAAADDCCRDAAAA;DAAAADDRDA;$$

A description number is

3133253117311335311173111332253111173111133517

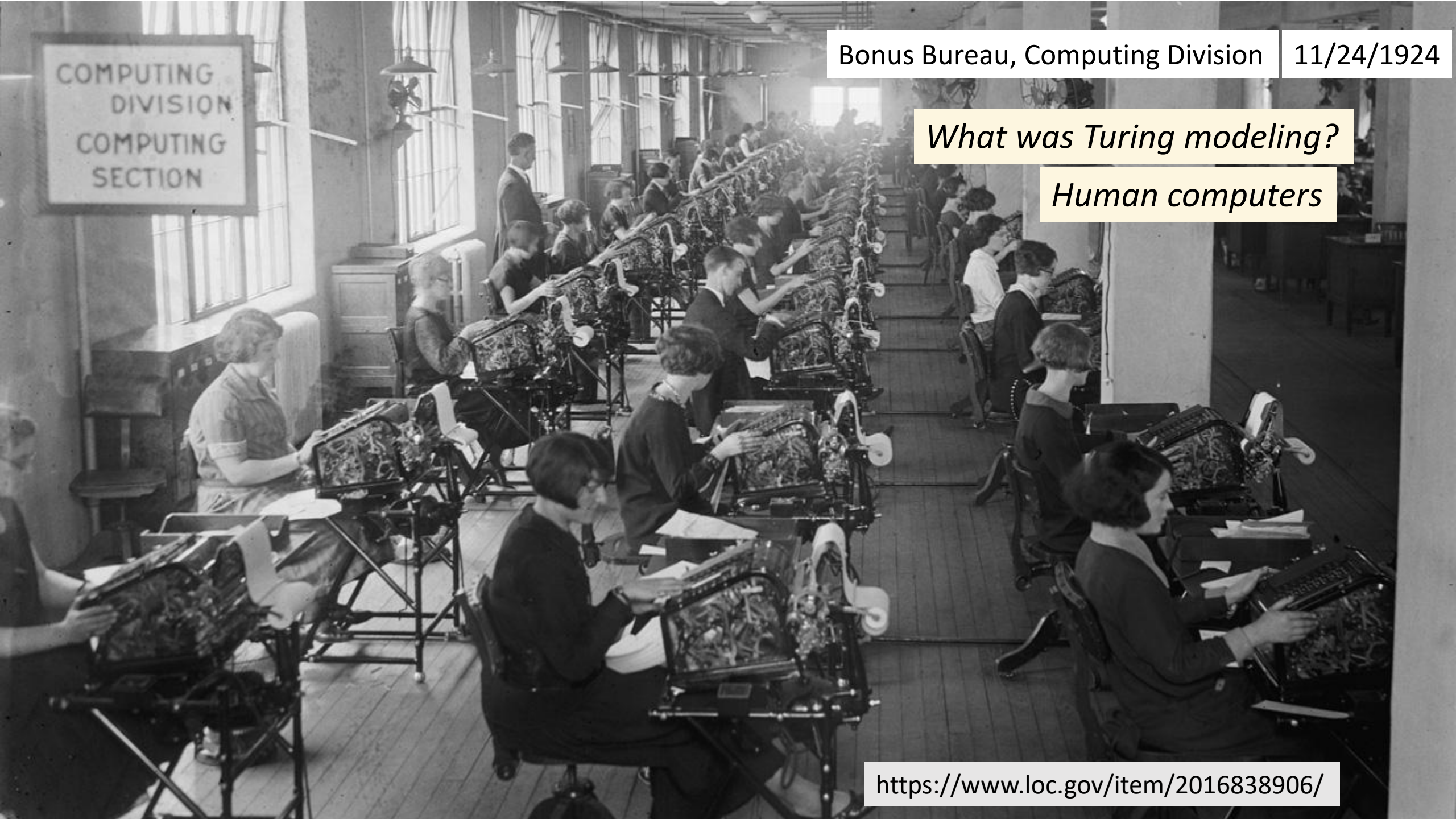
and so is

3133253117311335311173111332253111173111133531731323253117

Configuration		*Behaviour*	
m-config.	*symbol*	*operations*	*final m-config.*
$\mathfrak{b}$	None	$P0$, R	$\mathfrak{c}$
$\mathfrak{c}$	None	R	$\mathfrak{c}$
$\mathfrak{e}$	None	$P1$, R	$\mathfrak{f}$
$\mathfrak{f}$	None	R	$\mathfrak{b}$

Warning: Turning uses “m-config” to mean something different from modern usage.

Computing is normally done by writing certain symbols on paper. We may suppose this paper is divided into squares like a child's arithmetic book. In elementary arithmetic the two-dimensional character of the paper is sometimes used. But such a use is always avoidable, and I think that it will be agreed that the two-dimensional character of paper is no essential of computation. I assume then that the computation is carried out on one-dimensional paper, *i.e.* on a tape divided into squares. I shall also



Bonus Bureau, Computing Division | 11/24/1924

What was Turing modeling?

Human computers

https://www.loc.gov/item/2016838906/

Why model (human) computers?

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The "computable" numbers may be described briefly as the real
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Example of TM Programs

Palindrome: String that is same after reverse

Not regular.

TM can decide it.

https://morphett.info/turing/turing.html

```
; This example program checks if the input string is a binary palindrome.
; Input: a string of 0's and 1's, eg '1001001'


; Machine starts in state 0.

; State 0: read the leftmost symbol
0 0 _ r 1o
0 1 _ r 1i
0 _ _ * accept     ; Empty input

; State 1o, 1i: find the rightmost symbol
1o _ _ l 2o
1o * * r 1o

1i _ _ l 2i
1i * * r 1i

; State 2o, 2i: check if the rightmost symbol matches the most recently read left-hand symbol
2o 0 _ l 3
2o _ _ * accept
2o * * * reject

2i 1 _ l 3
2i _ _ * accept
2i * * * reject

; State 3, 4: return to left end of remaining input
3 _ _ * accept
3 * * l 4
4 * * l 4
4 _ _ r 0  ; Back to the beginning

accept * : r accept2
accept2 * ) * halt-accept

reject _ : r reject2
reject * _ l reject
reject2 * ( * halt-reject
```

Logistics

Next week:
three students explain their HW solutions.

You pick a question from the most recent HW.

No “bad” explanation.

Volunteer for 1% bonus:

Plan

Turing machine

Computability

Textbook [TCS] Section 7

https://introtcs.org/public/lec_06_loops.html
Loops and infinity

HW 5 due this Friday, Mar 13.
HW 6 coming soon.

Quiz 7 due next Monday
Anonymous mid-semester survey.